

Xiaolin Li

zimuxiaolin@gmail.com | marvin-huoshan.github.io | GitHub

RESEARCH FOCUS

AI security and privacy; robust watermarking and provenance for generative language, speech, and audio; adversarial machine learning; privacy-preserving learning under poisoning and backdoor attacks.

EDUCATION

Purdue University <i>Ph.D. in Computer and Information Technology</i>	Sep. 2023–Present West Lafayette, IN
Fujian Normal University <i>M.S. in Cybersecurity</i>	Sep. 2020–Jun. 2023 Fuzhou, China
Shandong Agricultural University <i>B.E. in Network Engineering</i>	Sep. 2016–Jun. 2020 Taian, China

SELECTED PUBLICATIONS

Xiaolin Li, Ning Wang, Ninghui Li, Wenhai Sun. Your Privacy My Cloak: Backdoor Attacks on Differentially Private Federated Learning. **IEEE Symposium on Security and Privacy (S&P), 2027. Accepted.**

Xiaolin Li, Ninghui Li, Boyang Wang, Wenhai Sun. Mitigating Data Poisoning Attacks to Local Differential Privacy. **ACM Conference on Computer and Communications Security (CCS), 2025.**

Xiaolin Li, Li Xu, Hongyan Zhang, Qikui Xu. Differential Privacy Preservation for Graph Auto-Encoders: A Novel Anonymous Graph Publishing Model. *Neurocomputing*, 2023.

Xiaolin Li, Qikui Xu, Zhenyu Xu, Hongyan Zhang, Li Xu. Graph Reconfigurable Pooling for Graph Representation Learning. *IEEE Transactions on Emerging Topics in Computing*, 2023.

RESEARCH EXPERIENCE

Machine Learning Engineer Intern <i>Xmotors AI</i>	Aug. 2026–Present Santa Clara, CA
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- Research audio watermarking and provenance for generative speech and audio, including robust detection after audio re-encoding.
- Develop evaluation pipelines on open speech and music models to assess compression, time shifts, and speed changes using multiple watermark keys, calibrated thresholds, and audio-quality metrics.

Research Assistant <i>Purdue University</i>	Sep. 2023–Present West Lafayette, IN
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- **RING (S&P 2027)**: Study how differential privacy can conceal malicious updates and enable backdoor attacks in federated learning.
- **LDP poisoning mitigation (CCS 2025)**: Develop multi-granularity detection and attack-resilient post-processing to recover utility in private frequency estimation.

Research Assistant <i>Fujian Normal University</i>	Sep. 2020–Jun. 2023 Fuzhou, China
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- Studied differential privacy and graph learning for privacy-preserving graph publication and representation learning.

HONORS AND PROFESSIONAL ACTIVITIES

Honors: RSAC 2026 Security Scholar (selected October 2025); Presidential Doctoral Excellence Award, Purdue University (2023).

Reviewer: IEEE INFOCOM 2026; IEEE TIFS 2025; IEEE TDSC 2025/2026; IEEE TKDE 2026; ACM TKDD 2025/2026; Journal of the Chinese Institute of Engineers 2025; The Journal of Supercomputing 2024.